

1.0 A Output Current, Dual Power Operational Amplifiers

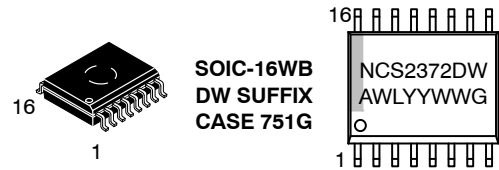
NCS2372

The NCS2372 is a monolithic circuit intended for use as a power operational amplifier in a wide range of applications, including servo amplifiers and power supplies. No deadband crossover distortion provides better performance for driving coils.

Features

- Output Current to 1.0 A
- Slew Rate of 1.3 V/μs
- Wide Bandwidth of 1.1 MHz
- Internal Thermal Shutdown
- Single or Split Supply Operation
- Excellent Gain and Phase Margins
- Common Mode Input Includes Ground
- Zero Deadband Crossover Distortion
- These Devices are Pb-Free and are RoHS Compliant

MARKING DIAGRAM



A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb-Free Package

PIN CONNECTIONS

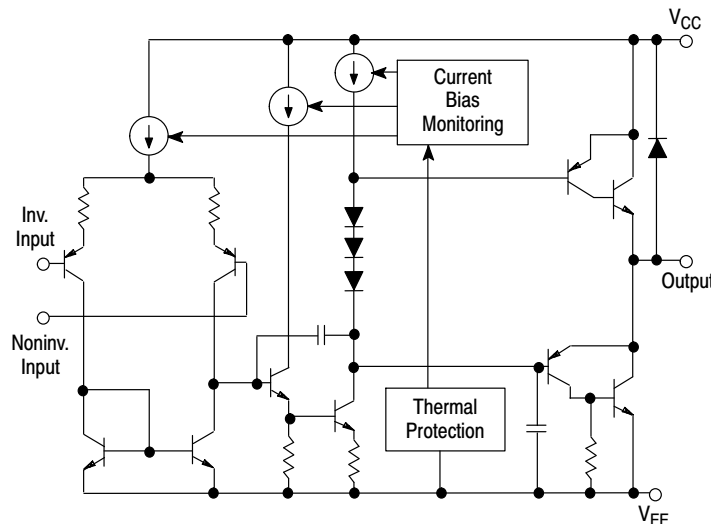
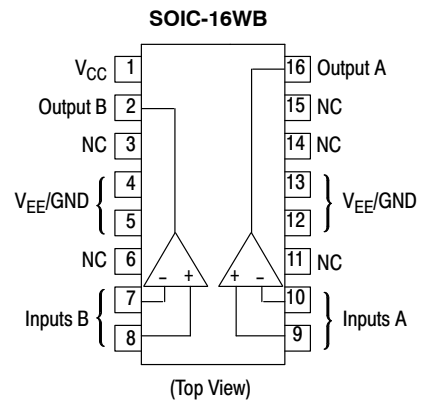


Figure 1. Representative Block Diagram

ORDERING INFORMATION

Device	Package	Shipping [†]
NCS2372DWR2G	SOIC-16WB (Pb-Free)	1000/Tape & Reel

[†] For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, [BRD8011/D](#).

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage (from V_{CC} to V_{EE})	V_S	40	V
Input Differential Voltage Range	V_{IDR}	Note 1	V
Input Voltage Range	V_{IR}	Note 1	V
Junction Temperature (Note 2)	T_J	+150	°C
Operating Temperature Range	T_A	-40 to +125	°C
Storage Temperature Range	T_{stg}	-55 to +150	°C
DC Output Current	I_O	1.0	A
Peak Output Current (Nonrepetitive) > 1 ms Duration < 1 ms Duration (Note 3)	$I_{(max)}$	1.5 2.0	A
Thermal Resistance, Junction-to-Air	$R_{\theta JA}$	80	°C/W
Thermal Resistance, Junction-to-Case	$R_{\theta JC}$	12	°C/W

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected. Operating the device outside its recommended conditions, but still within its maximum rated limits may not cause immediate damage. However, doing so can lead to reduced performance, unpredictable behavior, and potentially shorten the device's lifespan or reliability.

1. Either or both input voltages should not exceed the magnitude of V_{CC} or V_{EE} .
2. Power dissipation must be considered to ensure maximum junction temperature (T_J) is not exceeded.
3. When driving inductive loads, negative flyback voltage/current excursions may need to be constrained with Schottky diodes to protect the output drivers.



DC ELECTRICAL CHARACTERISTICS ($V_{CC} = +15\text{ V}$, $V_{EE} = -15\text{ V}$, R_L connected to ground, $T_A = -40^\circ$ to $+125^\circ\text{C}$.)

Characteristics	Symbol	Min	Typ	Max	Unit
Input Offset Voltage ($V_{CM} = 0$) $T_A = +25^\circ\text{C}$ T_A, T_{low} to T_{high}	V_{IO}	– –	1.0 –	15 20	mV
Average Temperature Coefficient of Offset Voltage	$\Delta V_{IO}/\Delta T$	–	20	–	$\mu\text{V}/^\circ\text{C}$
Input Bias Current ($V_{CM} = 0$)	I_{IB}	–	100	500	nA
Input Offset Current ($V_{CM} = 0$)	I_{IO}	–	10	50	nA
Large Signal Voltage Gain $V_O = \pm 10\text{ V}$, $R_L = 2.0\text{ k}$	A_{VOL}	30	100	–	V/mV
Output Voltage Swing ($I_L = 100\text{ mA}$) $T_A = +25^\circ\text{C}$ $T_A = T_{low}$ to T_{high} $T_A = +25^\circ\text{C}$ $T_A = T_{low}$ to T_{high}	V_{OH} V_{OL}	14.0 13.9 – –	14.2 – –14.2 –	– – –14.0 –13.9	V
Output Voltage Swing ($I_L = 1.0\text{ A}$) $V_{CC} = +24\text{ V}$, $V_{EE} = 0\text{ V}$, $T_A = +25^\circ\text{C}$ $V_{CC} = +24\text{ V}$, $V_{EE} = 0\text{ V}$, $T_A = T_{low}$ to T_{high} $V_{CC} = +24\text{ V}$, $V_{EE} = 0\text{ V}$, $T_A = +25^\circ\text{C}$ $V_{CC} = +24\text{ V}$, $V_{EE} = 0\text{ V}$, $T_A = T_{low}$ to T_{high}	V_{OH} V_{OL}	22.5 22.5 – –	22.7 – 1.3 –	– – 1.5 1.6	V
Input Common Mode Voltage Range $T_A = +25^\circ\text{C}$ $T_A = T_{low}$ to T_{high}	V_{ICR}	V_{EE} to $(V_{CC} - 1.0)$ V_{EE} to $(V_{CC} - 1.3)$			V
Common Mode Rejection Ratio ($R_S = 10\text{ k}$)	CMRR	70	90	–	dB
Power Supply Rejection Ratio ($R_S = 100\ \Omega$)	PSRR	70	90	–	dB
Power Supply Current $T_A = +25^\circ\text{C}$ $T_A = T_{low}$ to T_{high}	I_D	– –	8.0 –	10 14	mA

AC ELECTRICAL CHARACTERISTICS ($V_{CC} = +15\text{ V}$, $V_{EE} = -15\text{ V}$, R_L connected to ground, $T_A = +25^\circ\text{C}$, unless otherwise noted.)

Characteristics	Symbol	Min	Typ	Max	Unit
Slew Rate ($V_{in} = -10\text{ V}$ to $+10\text{ V}$, $R_L = 2.0\text{ k}$, $C_L = 100\text{ pF}$) $A_V = -1.0$, $T_A = T_{low}$ to T_{high}	SR	1.0	1.4	–	V/ μs
Gain Bandwidth Product ($f = 100\text{ kHz}$, $C_L = 100\text{ pF}$, $R_L = 2.0\text{ k}$) $T_A = 25^\circ\text{C}$ $T_A = T_{low}$ to T_{high}	GBW	0.9 0.7	1.4 –	– –	MHz
Phase Margin $T_J = T_{low}$ to T_{high} $R_L = 2.0\text{ k}$, $C_L = 100\text{ pF}$	ϕ_m	–	65	–	Degrees
Gain Margin $R_L = 2.0\text{ k}$, $C_L = 100\text{ pF}$	A_m	–	15	–	dB
Equivalent Input Noise Voltage $R_S = 100\ \Omega$, $f = 1.0$ to 100 kHz	e_n	–	22	–	$\text{nV}/\sqrt{\text{Hz}}$
Total Harmonic Distortion $A_V = -1.0$, $R_L = 50\ \Omega$, $V_O = 0.5\text{ VRMS}$, $f = 1.0\text{ kHz}$	THD	–	0.02	–	%

NOTE: In case V_{EE} is disconnected before V_{CC} , a diode between V_{EE} and Ground is recommended to avoid damaging the device.

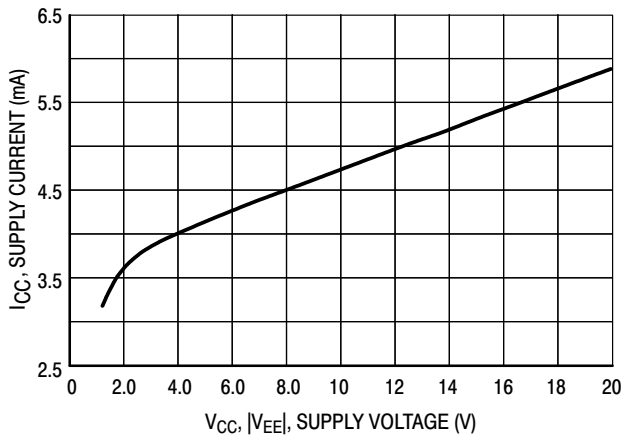


Figure 2. Supply Current versus Supply Voltage with No Load

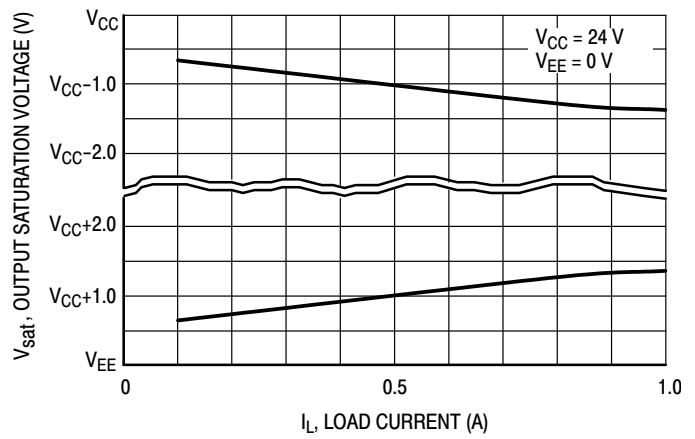


Figure 3. Output Saturation Voltage versus Load Current

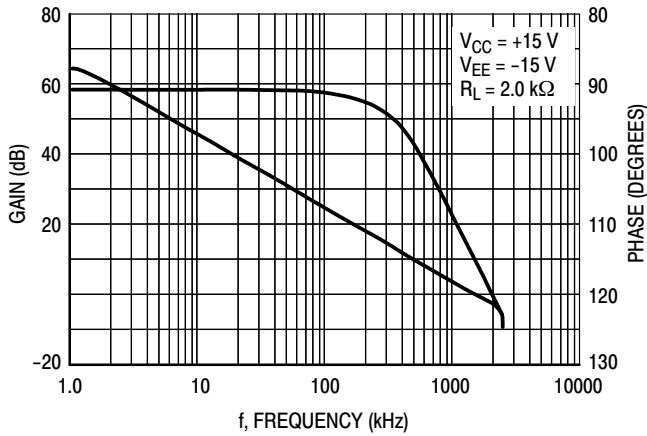


Figure 4. Voltage Gain and Phase versus Frequency

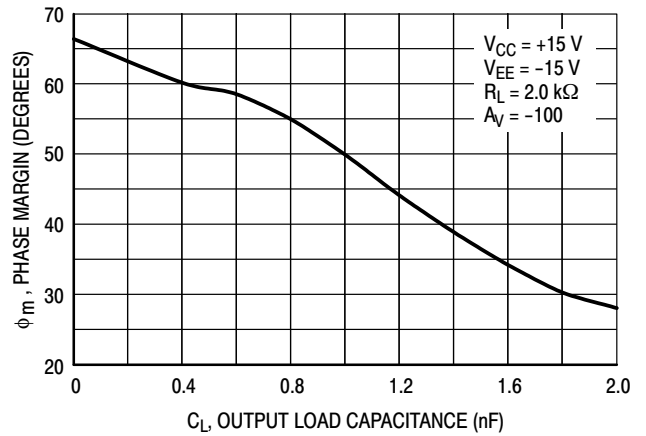


Figure 5. Phase Margin versus Output Load Capacitance

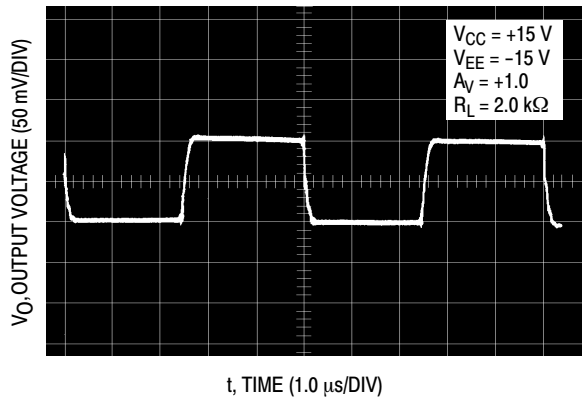


Figure 6. Small Signal Transient Response

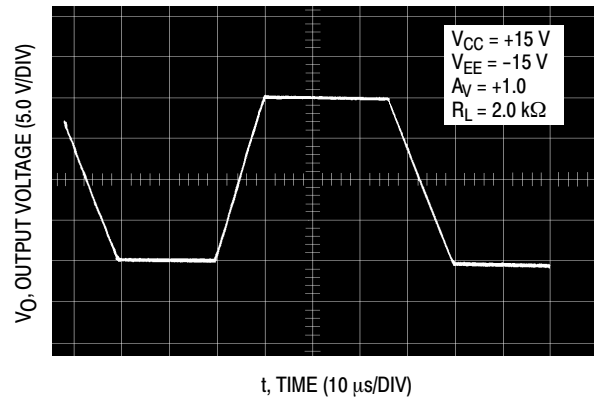


Figure 7. Large Signal Transient Response

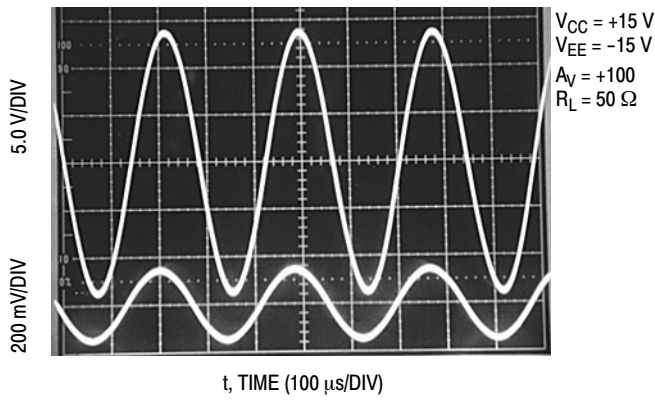


Figure 8. Sine Wave Response

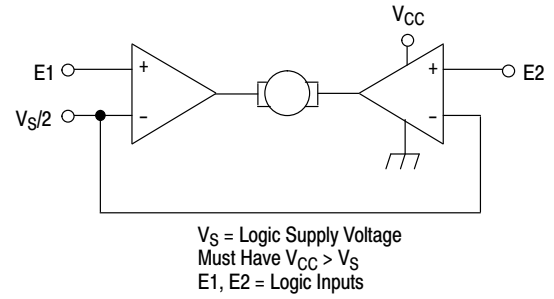
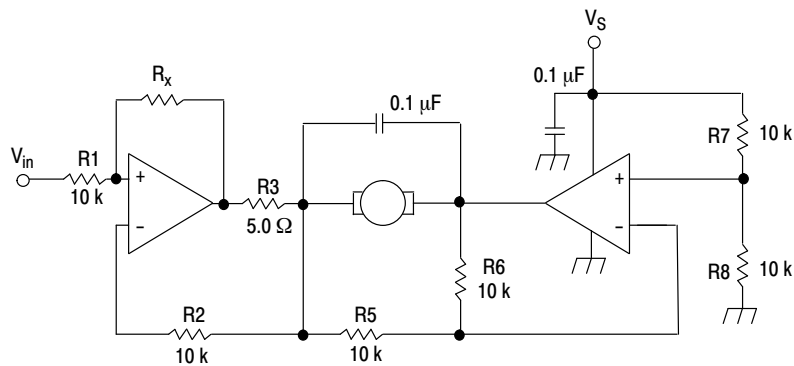


Figure 9. Bidirectional DC Motor Control with Microprocessor-Compatible Inputs



For circuit stability, ensure that $R_x > \frac{2R_3 \cdot R_1}{R_M}$ where, R_M = internal resistance of motor.
The voltage available at the terminals of the motor is: $V_M = 2(V_1 - \frac{V_S}{2}) + |R_0| \cdot I_M$
where, $|R_0| = \frac{2R_3 \cdot R_1}{R_x}$ and I_M is the motor current.

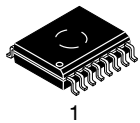
Figure 10. Bidirectional Speed Control of DC Motors

REVISION HISTORY

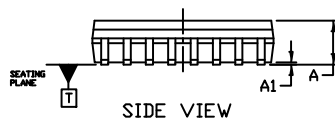
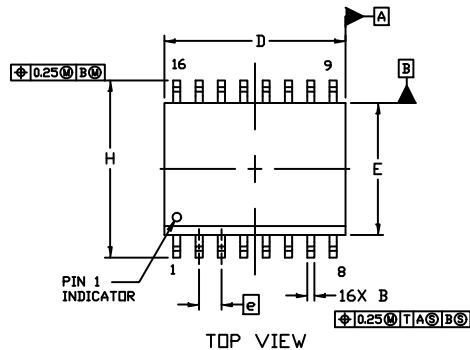
Revision	Description of Changes	Date
2	Rebranded the Data Sheet to onsemi format.	2/9/2026

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

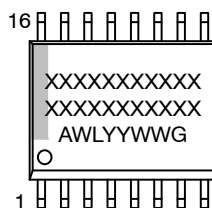




SCALE 1:1



GENERIC
MARKING DIAGRAM*



XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb-Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "•", may or may not be present. Some products may not follow the Generic Marking.

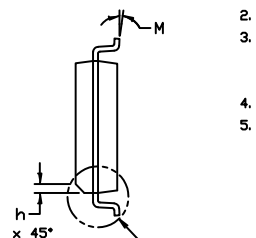
SOIC-16 WB
CASE 751G
ISSUE E

DATE 08 OCT 2021

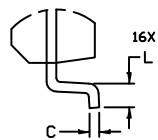
NOTES:

- DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
- CONTROLLING DIMENSION: MILLIMETERS
- DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE PROTRUSION SHALL BE 0.13 TOTAL IN EXCESS OF B DIMENSION AT MAXIMUM MATERIAL CONDITION.
- DIMENSIONS D AND E DO NOT INCLUDE MOLD PROTRUSIONS.
- MAXIMUM MOLD PROTRUSION OR FLASH TO BE 0.15 PER SIDE.

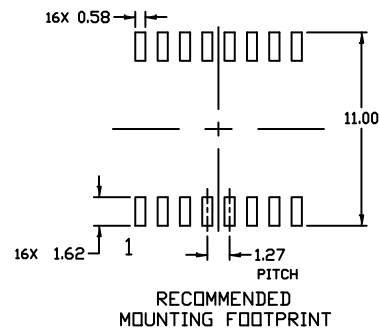
DIM	MILLIMETERS	
	MIN.	MAX.
A	2.35	2.65
A1	0.10	0.25
B	0.35	0.49
C	0.23	0.32
D	10.15	10.45
E	7.40	7.60
e	1.27	BSC
H	10.05	10.55
h	0.53	REF
L	0.50	0.90
M	0°	7°



END VIEW



DETAIL A
2X SCALE



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DESCRIPTION: SOIC-16 WB

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